Lecture 17: Taxes in the Q Theory

Juan Herreño
UCSD

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Tax Policy in the Q Theory

- Taxation of firms takes many forms
- Tax policy can, in principle, be used as countercyclical policy
- Examples of taxes/subsidies used
  - Business profit taxes
  - Dividend taxes
  - Investment tax credits: Deduct a share of investment expenditures
  - Bonus depreciation: Increases the speed at which firms can deduct investment expenditures

Bonus depreciation

TABLE 1—REGULAR AND BONUS DEPRECIATION SCHEDULES FOR FIVE-YEAR ITEMS

Year:	0	1	2	3	4	5	Total
Normal depreciation							
Deductions (000s)	200	320	192	115	115	58	1,000
Tax benefit (τ = 35 percent)	70	112	67.2	40.3	40.3	20.2	350
Bonus depreciation (50 percent)							
Deductions (000s)	600	160	96	57.5	57.5	29	1,000
Tax benefit (τ = 35 percent)	210	56	33.6	20.2	20.2	10	350

Source: Zwick and Mahon (2017)

Why bonus depreciation matters?

Summers (1987, p. 29.5) states this most clearly: *It is only because of discounting that depreciation schedules affect investment decisions"*

- Standard PV of deductions

$$z = \sum_{j=1}^{R} \frac{D_j^m}{(1+\pi)^j (1+r)^j}$$

- Bonus depreciation

$$\lambda_t + (1 - \lambda_t) z$$

- PV of tax benefits due to bonus depreciation

$$\zeta_t = (1 - \tau_d)\tau_\pi(\lambda_t + (1 - \lambda_t) z)$$

Naive Model

- Most of the theory you know is designed to understand non-durable goods
- Capital is durable (not only capital. Durable consumption is an important part of household expenditures)
- Standard supply and demand analysis may be arbitrarily uninformative if it ignores durability
- Let’s analyze that through an example

Naive Model

The world lasts for one period

- A firm that operates the production technology

$$y_t = k_t^{\alpha}$$

- I am assuming the firm only uses capital for simplicity
- Firm profits are

$$\pi_t = y_t - q_t(1 - \zeta_t)k_t$$

- where q_t is the relative price of capital
- Profit maximization yields the condition

$$k_t = \left(\frac{q_t(1 - \zeta_t)}{\alpha}\right)^{-1/(1-\alpha)}$$

- This is the capital demand equation.

What is the slope of capital demand?

In this interpretation, the capital demand elasticity $\varepsilon^d = 1/(1-\alpha)$ should be between 5 and 10. Why?

- At the optimal capital demand, the share of profits to output is

$$\frac{\pi}{y} = 1 - \alpha$$

- Make sure you derive this at home.
- So if the profit share of income is between 10% to 20%, then the elasticity of capital demand is between 5 and 10.

Capital supply and equilibrium

Assume a very simple capital supply equation

$$k_t = q_t^{\xi}$$

- Equate demand and supply and find the following equilibrium relation

$$k_t = \Theta(1 - \zeta_t)^{-\frac{\xi}{(1-\alpha)\xi+1}}$$

- For an uninteresting constant Θ
- Take logs and do a first-order Taylor expansion around a point where $\zeta_t = \bar{\zeta}$

$$\log k_t \approx \log \bar{k} + \frac{\xi}{(1-\alpha)\xi + 1} \frac{1}{1-\bar{\zeta}} (\zeta_t - \bar{\zeta})$$

What is the value of ξ

Idea! Use knowledge on the profit share and the causal effects of ζ on k to back out ξ.

- Rearrange using $\zeta = z \times \tau$

$$\Delta \log k_t \approx \frac{\xi}{(1-\alpha)\xi + 1} \frac{\tau}{1 - \bar{z}\tau} \Delta z_t$$

- Use the definition of supply and demand elasticities

$$\Delta \log k_t \approx \frac{\varepsilon^s \varepsilon^d}{\varepsilon^s + \varepsilon^d} \frac{\tau}{1 - \bar{z}\tau} \Delta z_t$$

- Zwick and Mahon '17 (on your reading list for next class) uses $\Delta z \approx 0.05, \bar{z} \approx 0.9, \tau = 0.35, \Delta \log k_t \approx 0.17$. Therefore:

$$\varepsilon^s \approx \frac{-7\varepsilon^d}{7 - \varepsilon^d}$$

What is the value of ξ

If $\alpha = 0.8$ such that $\varepsilon = 5$.

$$\varepsilon^S \approx \frac{-35}{2}$$

- A negative supply elasticity?!

If $\alpha = 0.9$ such that $\varepsilon = 10$.

$$\varepsilon^S \approx 23$$

- Very large supply elasticity?

Apparently, little disagreement about the profit share changes inference by a lot!

Should the profit share dictate the capital demand elasticity?

- In this simple model, the shape of the production function, and nothing else, dictates the capital demand elasticity
- Sensible assumptions when firms are buying/renting non-durable inputs
- But firms are not supposed to maximize today's profits, but the PV of profits, and buying capital today has implications over those future profits.
- Let's see what a model with durable inputs has to say about the shape of the capital demand equation

Firm problem with taxes

$$V_t = \max \sum_{j=0}^{\infty} \Lambda_{t,t+j} \left(F(K_{t+j})(1-\tau^{\pi})(1-\tau^{d}) - I_t p_t (1-\zeta_t) \right)$$

subject to:

$$K_{t+1} = K_t(1-\delta) + I_t$$

- Simplifying assumption: Firm internalizes the PV of the deductions at time t. Common in key references, e.g. Hall and Jorgenson (1967), House and Shapiro (2008)
- No labor explicitly. Easy to extend if labor market is spot, just more notation
- Firm buys capital from outside producer at price p. Adjustment costs occur on that capital producing firm. Referred in the literature as *external* adjustment costs

Optimality conditions

$$q_t = p_t(1 - \zeta_t)$$

$$q_t = q_{t+1}\Lambda_{t,t+1}(1 - \delta) + \Lambda_{t,t+1}\left(F_k(K_{t+1})(1 - \tau^{\pi})(1 - \tau^{d})\right)$$

- Question: What is the effect of a change in ζ_t?
- Slightly simpler version of House and Shapiro (2008)
- Pedagogical exercise: imagine a transient policy
- The change in ζ does not last for long
- Assume capital is highly durable ($\delta \approx 0$)
- And firms are very forward looking ($\beta \approx 1$)

Transitory Policy

Let me assume that the change in ζ is transitory. $\zeta_{t+1} = \bar{\zeta}$. So the economy tomorrow will be close to the steady state.

- $q_{t+1} \approx q^{ss}$
- $\Lambda_{t,t+1} \approx \beta$
- δ is small so that $\delta^2 \approx 0$ (For structures $\delta^2 = 0.0004$)

Transform:

$$q_t = q_{t+1}\Lambda_{t,t+1}(1-\delta) + \Lambda_{t,t+1}\left(F_k(K_{t+1})(1-\tau^{\pi})(1-\tau^{d})\right)$$

into:

$$q_t = q^{ss}\beta(1-\delta) + \beta\left(F_k(K_{t+1})(1-\tau^{\pi})(1-\tau^{d})\right)$$

Transitory Policy

$$q_t = q^{ss}\beta(1-\delta) + \beta\left(F_k(K_{t+1})(1-\tau^{\pi})(1-\tau^{d})\right)$$

In steady state:

$$q^{ss} = \frac{\beta}{1-\beta(1-\delta)}\left(F_k(K^{ss})(1-\tau^{\pi})(1-\tau^{d})\right)$$

Divide the first equation by the second one and rearrange

$$q_t \approx q^{ss}\left(\beta(1-\delta) + (1-\beta(1-\delta))\frac{F_k(K_{t+1})}{F_k(K^{ss})}\right)$$

- For structures $1-\beta(1-\delta) = 0.052$
- $K/Y = 4$, and $I/Y = 0.18$, so $I/K = 0.045$
- Imagine the reform had a large effect on investment. Second term still very small.

Economics

- On one side $q_t \approx q^{SS}$
  - Capital is long lived. So the marginal benefit is dominated by stream of future MPKs into the distant future. Those do not change much with a temporary policy
- On the other side

$$q_t = p_t(1 - \zeta_t)$$

- the after-tax price of investment must be constant!
- If investment subsidies go up, the pre-tax price must go up.
- The demand for capital does not depend on I_t. It is horizontal.
- If capital is perfectly durable, firms are infinitely elastic on when to invest. If capital becomes cheaper today than tomorrow, I will invest today **rather than** tomorrow.
- Note this argument is about the **timing** of investment expenditures, rather than the optimal size of the firm (which is given by steady state parameter values).

With internal as opposed to external adjustment costs

When adjustment costs are internal, the firm is the capital producer

- The firm will internalize that the effective cost of investment rises with the size of investment
- The firm internalizes that investing more today reduces adjustment costs in the future

This is not an *either or* question. In principle firms may buy capital from outside firms with increasing marginal costs, and face installation costs

Firm problem with taxes

$$V_t = \max \sum_{j=0}^{\infty} \Lambda_{t,t+j} \left(F(K_{t+j})(1-\tau^{\pi})(1-\tau^{d}) - I_t(1-\zeta_t) - \frac{\varphi}{2}\left(\frac{I_{t+j}}{K_{t+j}}\right)^2 K_{t+j} \right)$$

subject to:

$$K_{t+1} = K_t(1-\delta) + I_t$$

Optimality conditions

$$\frac{I_t}{K_t} = \frac{1}{\varphi}(q_t - (1 - \zeta_t))$$

$$q_t = q_{t+1}\Lambda_{t,t+1}(1-\delta) + \Lambda_{t,t+1}\left(F_k(K_{t+1})(1-\tau^{\pi})(1-\tau^{d}) + \frac{\varphi}{2}\left(\frac{I_{t+1}}{K_{t+1}}\right)^2\right)$$

Economics

- On one side $q_t \approx q^{ss}$
  - Capital is long lived. So the marginal benefit is dominated by stream of future MPKs into the distant future. Those do not change much with a temporary policy
- On the other side

$$\frac{I_t}{K_t} = \frac{1}{\varphi}(q_t - (1 - \zeta_t))$$

  - The relative price of investment is decreasing from $(1 - \zeta)$ to $(1 - \zeta_t)$ when ζ increases
  - It must be that firms are investing more!

$$\frac{dI/K}{d\zeta} = \frac{1}{\varphi}$$

  - Given our argument about $dq/d\zeta \approx 0$

Economics

Notice a couple of insights

- The reaction of investment to temporary tax policy depends critically on φ
- The argument relies on a very horizontal demand curve for investment
  - The marginal value of capital depend on stream of future MPKs
  - Changes in the after-tax relative price of investment in one period will convince firms to change the timing of their investments
- $\frac{\partial I/K}{\partial q} = 1/\varphi$
  - The responsiveness of I to ζ depends on how costly is to build-up capital. The slope of capital supply is given by the adjustment cost function

Numerical Results

Table 1—Response to a Temporary Investment Subsidy

Duration	Depreciation rate	Shadow price (φ) $\xi = 0$	$\xi = 0.5$	$\xi = 1$	$\xi = 5$	$\xi = 10$	$\xi = 15$	$\xi = 20$
6 months	$\delta = 0.001$	1.000	1.000	1.000	0.999	0.998	0.997	0.996
	$\delta = 0.01$	1.000	0.999	0.998	0.992	0.986	0.982	0.978
	$\delta = 0.02$	1.000	0.998	0.996	0.986	0.976	0.969	0.963
	$\delta = 0.05$	1.000	0.996	0.992	0.970	0.951	0.936	0.923
	$\delta = 0.10$	1.000	0.992	0.985	0.945	0.911	0.885	0.864
	$\delta = 0.25$	1.000	0.982	0.965	0.877	0.807	0.755	0.714
1 year	$\delta = 0.001$	1.000	1.000	0.999	0.997	0.995	0.993	0.991
	$\delta = 0.01$	1.000	0.998	0.996	0.983	0.972	0.964	0.956
	$\delta = 0.02$	1.000	0.996	0.993	0.972	0.954	0.940	0.928
	$\delta = 0.05$	1.000	0.992	0.984	0.941	0.906	0.878	0.855
	$\delta = 0.10$	1.000	0.985	0.971	0.896	0.835	0.790	0.753
	$\delta = 0.25$	1.000	0.966	0.936	0.784	0.673	0.597	0.539
2 years	$\delta = 0.001$	1.000	0.999	0.999	0.995	0.990	0.986	0.983
	$\delta = 0.01$	1.000	0.996	0.992	0.967	0.946	0.930	0.915
	$\delta = 0.02$	1.000	0.992	0.985	0.946	0.912	0.886	0.864
	$\delta = 0.05$	1.000	0.984	0.969	0.890	0.826	0.779	0.740
	$\delta = 0.10$	1.000	0.971	0.946	0.814	0.715	0.645	0.591
	$\delta = 0.25$	1.000	0.941	0.891	0.659	0.515	0.428	0.368
3 years	$\delta = 0.001$	1.000	0.999	0.998	0.992	0.985	0.980	0.975
	$\delta = 0.01$	1.000	0.993	0.988	0.952	0.922	0.898	0.878
	$\delta = 0.02$	1.000	0.989	0.979	0.921	0.873	0.837	0.807
	$\delta = 0.05$	1.000	0.976	0.956	0.845	0.760	0.698	0.649
	$\delta = 0.10$	1.000	0.959	0.925	0.749	0.626	0.545	0.485
	$\delta = 0.25$	1.000	0.922	0.860	0.587	0.439	0.357	0.304
Permanent	$\delta = 0.001$	1.000	0.986	0.972	0.884	0.806	0.749	0.704
	$\delta = 0.01$	1.000	0.929	0.872	0.637	0.513	0.443	0.396
	$\delta = 0.02$	1.000	0.908	0.839	0.578	0.453	0.387	0.343
	$\delta = 0.05$	1.000	0.888	0.808	0.528	0.405	0.341	0.300
	$\delta = 0.10$	1.000	0.879	0.794	0.506	0.384	0.322	0.282
	$\delta = 0.25$	1.000	0.872	0.783	0.489	0.367	0.306	0.267

Notes: The table shows the equilibrium percent change in the shadow price of capital goods φ in response to an investment subsidy of 1 percent ($d\xi = 0.01$). Investment supply is given by equation (5). For the numerical calculations, the production function is $AK_t^{\alpha}F_t^{\alpha}$, $r = 0.02$, and $\alpha = 0.35$.

Source: House and Shapiro (2008). The effect on q is 1 minus the effect on φ (which in their notation is the shadow price of investment), and $\xi = (\delta\varphi)^{-1}$ in our notation.